Trần Quốc Anh N26D ECE077

P1: 2,25 / 2,25 Problem 1:

P2: 2,5

P3: 1,0

P4: 2,0

7,75

Given a singly linked list:

head → [10] → [25] → [40] → [15] → NULL

1. Write a function pseudocode to reverse the singly linked list, including three pointers (prev, curr, next) to complete your function

2. Draw a diagram or step-to-step pointer trace table showing the pos and values of three pointers

2,5 Problem 2:

Given a singly circular linked list:

head → [10] → [20] → [30] → head

1. Write pseudocode InsertHead(head, newValue) to insert a new node containing newValue at the front of the circular linked list while maintaining its circular structure.

2. Trace the execution steps when inserting newValue = 5 into the given list.

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Problem 3:

Given a doubly linked list containing int.

1. Write pseudocode for InsertBefore(head, target, newValue)

2. list the order of pointer assignments required and briefly justify why this avoid memory leak or broken links.

3. Explicitly handle the following in pseudocode:

a) Target is located at head node.

b) _____ does not exist

Problem 4:

head → [2] → [5] → [8] → NULL.

Task 1: write SortInsert(head, val)

2: Construct step-to-step trace table detailing pointer when inserting val = 6 into the list.

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~~Test~~

Problem 1.

1,25 Pseudocode :

```
function reverseLinkedList (head) {
    prev = NULL
    next = NULL
    curr = head
    while (curr is not NULL :)
        next = curr.next
        curr.next = prev
        prev = curr
        curr = next
    head = prev
    return head.
}
```

~~0,7~~ Diagram :

1,00

iteration	Prev	curr	next	list State
initialization	NULL	[10]	NULL	No change
1	[10]	[25]	[25]	[10] → NULL
2	[25]	[40]	[40]	[25] → [10] → Null
3	[40]	[15]	[15]	[40] → [25] → [10] → [NULL]
4	[15]	NULL	NULL	[15] → [40] → [25] → [10] → Null.

Final result ?

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Problem 2:

1.25 Pseudocode:

```
function InsertHead(head, newValue){
    newNode = createNode(newValue)
    if head is Null{
        newNode.next = newNode
        return newNode
    }
    temp = head

    While temp.next != head {
        temp = temp.next
    }
    newNode.next = head
    temp.next = newNode
    return newNode
}
```

1.75 Trace newValue = 5:

1. Create newNode = 5.
2. Set temp to head ([10])
3. Traverse the list with temp until temp.next equals head. temp moves to [20] then to [30]. Loop end because node [30] points to head([10]).

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4. Set newNode.next = head so that [5] points to [10].

5. temp.next = newNode so that [30] points to new node [5].

6. return newNode as new head.

result is now:

head → [5] → [10] → [20] → [30] → head.

10

Problem 3.

Pseudocode:

```
function InsertBefore (head, target, newValue ){
  -newNode = createNode (newValue)
  // Edge case: List is Empty
  if head is NULL:
      return NULL

  // Target is Located at head node
  if head.value == target:
      newNode.prev = NULL
      newNode.next = head
      head.prev = newNode
      return newNode
```


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```
    curr = head
    // Traverse to find the target node
    While curr is not NULL and curr.value != target:
        curr = curr.next

    // Edge case: Target doesn't exist
    if curr is NULL:
        return head

    Node prev
    prevNode = curr.prev
    newNode.next = curr
    newNode.prev = prevNode
    prevNode.next = newNode
    curr.prev = newNode
    return head
}
```

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Problem 4:

Pseudo code:

```
function SortedInsert ( head, ~~val~~Value ):
    newNode = Value(val)
    if ( !head || ~~and~~ val < head -> data ):
        newNode -> Next = head
        return newNode

    curr = head
    while curr -> next && curr -> next -> data < va
        curr = curr -> next

    NewNode -> nex = curr -> next,
        curr -> next = newNode
        return head
```

~~Trace:~~

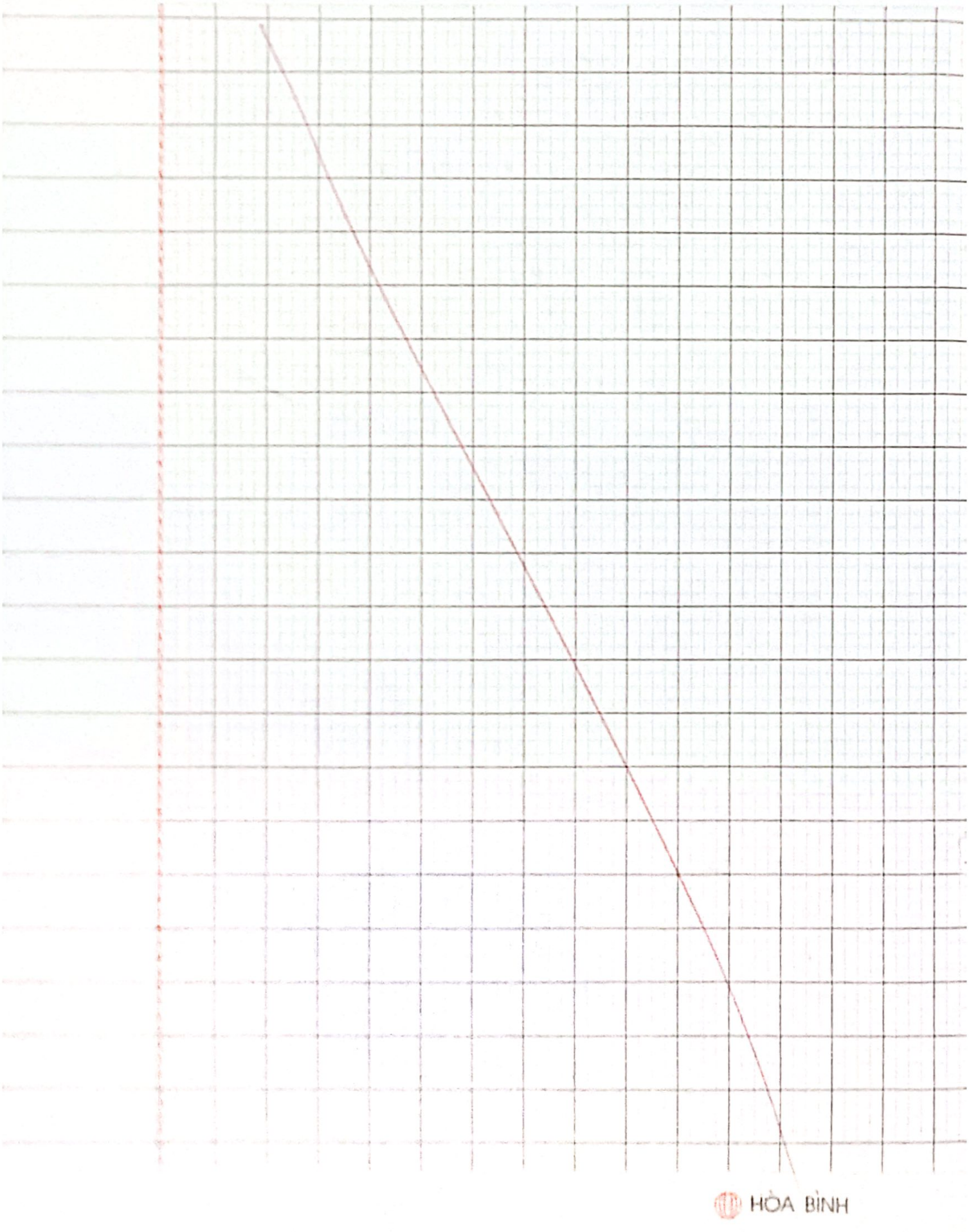
Trace:

Step	curr	Stat	Action
1	[2]	5 < 6 (true)	move to [5]
2	[5]	8 < 6 (false)	stop loop
3	[5]	None	link [5] -> [6] -> [8]

Final result ?

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